

Phase 4: Project Planning Phase

AI Brand Voice & Marketing Content Generator

1. Project Scope & MVP Definition

The scope of this project is to construct a fully functional Minimum Viable Product (MVP) of the AI Brand Voice Generator. This involves establishing Gemini LLM connectivity, building core database-backed logic for tone analysis, implementing a fluid Streamlit frontend, styling the interface with professional typography and CSS, and validating quality across various real-world brand test profiles.

2. Work Breakdown Structure (Epics & Stories)

Project tasks are divided into 5 development Epics, each detailing specific User Stories and expected physical deliverables:

WBS Epic	Associated Stories	Est. Effort	Physical Deliverable
Epic 1: Infrastructure	S1: Obtain Gemini API Key S2: Setup Environment & Parameters S3: Validate LLM Connectivity	3 Hours	Functional config, local .env files, and a terminal connectivity test script.
Epic 2: Core Logic	S1: Brand Style Learning Logic S2: Multi-Format Content Engine S3: Error Handling & Validations	5 Hours	Robust functions (learn_brand_voice, generate_brand_content) in database.py.
Epic 3: UI & App Logic	S1: Page Structure & Setup S2: Content Display Containers S3: Copy Button integration S4: Interactive Feedback Loop	6 Hours	Interactive Streamlit web app integrating backend logic and pyperclip utilities.
Epic 4: Enhancements	S1: Layout Design & Branding S2: Custom CSS Theme & Gradients S3: Spinner Indicators & Alerts S4: About & Usage Summary Section	4 Hours	Highly styled interface with custom CSS sheets (.feature-card hover, gradient header).
Epic 5: QA & Testing	S1: Professional Brand Inputs Test S2: Output Quality Validation Matrix S3: Latency & Performance Tuning	3 Hours	Comprehensive testing reports, latency profiles, and optimized query pipelines.

3. Execution Timeline

The total development cycle spans approximately 21 hours of focused implementation. Development follows a strict sequential pipeline, ensuring that core logical engines are completely finalized, database-backed, and verified before the frontend UI is bound and visual enhancements are integrated.